
Penalized Regression, Exercise

Name: _____ SBU ID: _____

Please include (1) Rmd file; (2) Output from Rmd with answers to all the questions asked.

Please refer to the following website for Penalized Regressions using R:

<http://www.sthda.com/english/articles/37-model-selection-essentials-in-r/153-penalized-regression-essentials-ridge-lasso-elastic-net/>

Penalized Regression with the Boston Housing Data

This dataset contains information collected by the U.S Census Service concerning housing in the area of Boston Mass. It was obtained from the StatLib archive (<http://lib.stat.cmu.edu/datasets/boston>), and has been used extensively throughout the literature to benchmark algorithms. The goal is to model the variable **MEDV** using the other 13 variables.

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|-------------|---|
| 1. CRIM | per capita crime rate by town |
| 2. ZN | proportion of residential land zoned for lots over 25,000 sq.ft. |
| 3. INDUS | proportion of non-retail business acres per town |
| 4. CHAS | Charles River dummy variable (= 1 if tract bounds river; 0 otherwise) |
| 5. NOX | nitric oxides concentration (parts per 10 million) |
| 6. RM | average number of rooms per dwelling |
| 7. AGE | proportion of owner-occupied units built prior to 1940 |
| 8. DIS | weighted distances to five Boston employment centres |
| 9. RAD | index of accessibility to radial highways |
| 10. TAX | full-value property-tax rate per \$10,000 |
| 11. PTRATIO | pupil-teacher ratio by town |
| 12. B | $1000(B_k - 0.63)^2$ where B_k is the proportion of blacks by town |
| 13. LSTAT | % lower status of the population |
| 14. MEDV | Median value of owner-occupied homes in \$1000's |

1. Please use the random seed 123 to divide the data into 75% training and 25% testing.
2. Please first find the best **Ridge Regression model** using **the training data**. Please (a) find the best λ value through cross-validation and display this value; (b) display the coefficients of the fitted model; and (c) make prediction on **the testing data**, and report the RMSE and the Coefficient of Determination R^2 .
3. Please first find the best **LASSO model** using **the training data**. Please (a) find the best λ value through cross-validation and display this value; (b) display the coefficients of the fitted model; and (c) make prediction on **the testing data**, and report the RMSE and the Coefficient of Determination R^2 .

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4. Please first find the best [Elastic Net model](#) using [the training data](#). Please (a) find the best **tuning parameter** values through cross-validation and display these values; (b) display the coefficients of the fitted model; and (c) make prediction on [the testing data](#), and report the RMSE and the Coefficient of Determination R^2 .

